

Data Science Capstone Project Proposal

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Proposed Title:	LIKAS: A Web-Based Decision Support System for Flood Rescue Prioritization Using Random Forest and XGBoost
CS Research Categories/Field:	Decision Support Systems, Machine Learning, Disaster Informatics, Web Application Development
TECHNICAL BACKGROUND: Background of the Study Flooding remains one of the most devastating and frequently occurring natural disasters in the Philippines, particularly in densely populated urban areas such as Metro Manila. The National Capital Region (NCR) has long been exposed to recurring flood events that cause significant loss of life, mass displacement, and widespread economic damage. In the context of barangay-level communities, managing flood preparedness and evacuation planning is particularly critical due to the concentration of vulnerable populations, including senior citizens, persons with disabilities (PWDs), young children, and pregnant women, in low-lying, flood-prone streets. Dulawan et al. [1] examined the social drivers of flood vulnerability among households in Metro Manila, highlighting how persistent exposure to flooding is deeply intertwined with socioeconomic marginalization and limited capacity for self-evacuation. Understanding the distribution of these vulnerabilities at the granular street level, and translating that understanding into structured data frameworks, is therefore essential for designing effective local disaster response systems. Vulnerability assessment frameworks have been widely recognized as a foundational component of effective disaster preparedness planning. In the Philippine context, barangay disaster risk reduction and management (DRRM) units are legally mandated under Republic Act No. 10121 to lead local disaster risk reduction efforts, including the identification of high-risk areas and the coordination of local safety activities [2]. However, in practice, these responsibilities are often carried out without access to systematic tools for recording and analyzing area-level vulnerability data across multiple barangays at once. The NDRRMC's situational reports on major typhoon events documented that thousands of families were displaced with limited	

prior coordination at the micro-level, reflecting the operational gaps that persist when preparedness planning relies on manual, experience-based judgment rather than structured, centralized data [3]. Investigating how socio-demographic indicators, such as population size, historical flood depth, and counts of at-risk residents per street, can be systematically aggregated across an entire city and used to train a predictive classification model offers a critical pathway toward more equitable and organized flood preparedness.

Recent developments in machine learning and data-driven classification have significantly advanced the field of disaster risk assessment, enabling more accurate and actionable prioritization of vulnerable areas. Liu et al. [4] demonstrated that ensemble-based algorithms, particularly Random Forest (RF) and XGBoost, can effectively integrate multiple flood risk and vulnerability indicators, including population density, proportion of highly sensitive demographic cohorts, and localized flood frequencies, into unified risk scoring and classification outputs. These ensemble models are particularly well-suited for barangay-level applications because of their robustness with heterogeneous input data, resistance to overfitting, and capacity to handle non-linear interactions across multiple vulnerability dimensions simultaneously. By pairing trained Random Forest and XGBoost classifiers, used comparatively for prediction accuracy, with an interpretable, formula-based scoring layer, this study aims to explore the performance, accuracy, and operational viability of applying machine learning-based priority level determination to support structured, data-driven evacuation planning at the local government level.

Building on these foundations, the developed system, LIKAS, is a web-based Decision Support System (DSS) built for the Manila City Disaster Risk Reduction and Management Office (MDRRMO) and its constituent Barangays. The system follows a hybrid scoring approach: interpretable formulas first compute a Vulnerability Score and a Hazard/Priority Score for a given street or barangay, and trained Random Forest and XGBoost classifiers then combine these engineered features with flood and demographic data to output a final rescue priority classification. LIKAS integrates a Street Registry and Population Vulnerability module for encoding street-level socio-demographic data, a Flood Incident Management module with an administrative approval workflow for verifying incoming flood reports, and an interactive Priority Ranking Dashboard with geospatial mapping and analytics. By systematically processing street-level vulnerability data through this hybrid engine into automated, high-confidence priority rankings, LIKAS shifts barangay-level flood preparedness from reactive, experience-based planning toward a proactive, data-driven approach, deployed city-wide rather than for a single barangay.

Statement of the Problem

STATEMENT OF THE PROBLEM:

General Problem:

Local Government Units (LGUs) and Barangay Disaster Risk Reduction and Management (DRRM) Officers currently lack a centralized, data-driven system for determining which flood-affected streets should be prioritized for rescue and response operations, resulting in decisions that rely heavily on manual judgment and incomplete information rather than a consistent, city-wide evaluation of flood severity and population vulnerability together.

Specifically, this study aims to answer the following:

1. How can flood severity and demographic vulnerability be analyzed together to accurately identify and rank which affected streets require the most urgent rescue response, instead of relying on incomplete or one-dimensional information?
2. How can a reliable, data-driven tool be developed to help emergency responders and LGU officials determine which flood-affected zones require immediate intervention and resource allocation, reducing their reliance on manual judgment and fragmented reports gathered separately from different barangays?
3. How can disconnected flood-related information, currently presented through printed maps, phone calls, and paper-based records, be consolidated into an accessible, user-friendly web platform with intuitive dashboards and interactive geospatial visualizations for faster comprehension during emergencies?
4. How can demographic vulnerability data, specifically the presence of persons with disabilities (PWDs), senior citizens, pregnant women, and children, be systematically incorporated into flood response evaluation to ensure that highly vulnerable populations are consistently prioritized for immediate rescue?
5. How can an analytical tool be designed to evaluate and compare flood impact and demographic vulnerability across multiple barangays simultaneously, providing city-wide (MDRRMO-level) visibility into which barangays are most in need of support during widespread flooding events?

HOW DID OTHERS SOLVE THE PROBLEM?

The traditional approach to localized flood management relies heavily on geospatial tools to simulate and visualize environmental threats. Local researchers and disaster risk units universally deploy Geographic Information Systems (GIS), such as Esri's ArcGIS or QGIS, to execute flood risk assessments and map geographic susceptibility. These models excel at evaluating complex terrain-related and environmental data, including elevation parameters, historic rainfall vectors, and hydrological characteristics, to establish exactly where flooding is bound to occur [5].

However, a critical gap persists between mapping a physical hazard zone and executing rapid, tactical rescue operations. Traditional flood risk frameworks are heavily biased toward environmental indicators and structural exposure, frequently

ignoring micro-scale social and demographic vulnerabilities [6]. As noted by academic literature, vulnerability analysis remains the weakest link in contemporary disaster mapping; while physical hazard and exposure modeling are technically advanced, indicator-based mapping at the micro-scale remains severely underdeveloped [7].

Because traditional GIS hazard maps function primarily on macro- or aggregated spatial levels, they fail to account for the unique socio-economic characteristics that dictate human survival during a sudden inundation. For instance, two adjacent households situated within the exact same flood depth zone will experience vastly different coping capacities based on their internal demographics. A standard GIS map illustrates the spatial boundaries of the water but cannot identify which specific street shelters immobile elderly individuals, infants, or pregnant women who require immediate evacuation assistance [6]. Consequently, relying solely on traditional environmental maps restricts local government units to a reactive, first-come-first-served rescue paradigm. To maximize the efficiency of localized tactical response, modern emergency decision support systems must actively synthesize physical risk boundaries with granular, street-level socio-demographic vulnerability indices [6], [8].

HOW DO YOU PLAN TO SOLVE THE PROBLEM?

The proponents developed LIKAS, a web-based Decision Support System (DSS) that addresses the absence of a centralized, data-driven tool for city-wide flood preparedness by integrating a street-level vulnerability registry, a hybrid formula-and-machine-learning priority classification engine, and a validated incident logging workflow into a single, unified platform serving the Manila City MDRRMO and all of its barangays.

LIKAS operates through three interconnected functional layers:

1. Data Collection and Registry Layer

Using a React frontend built with Vite (styled with Tailwind CSS), backed by a FastAPI (Python) REST API connected to a PostgreSQL database, LIKAS provides structured data interfaces organized around a district, city, and barangay hierarchy. Users authenticate securely using JSON Web Token (JWT) authentication with a 24-hour session expiry. Admin/MDRRMO accounts view data across the entire city, while Barangay accounts are automatically scoped to their own jurisdiction. Through the Street Registry and Population Vulnerability modules, users encode localized records capturing the street name, geographic coordinates, total population, and exact counts of high-risk residents: senior citizens, PWDs, children, and pregnant women. This structured encoding replaces informal tracking with a relational database, forming a standardized feature set for the downstream scoring engine.

2. Hybrid Scoring and Priority Classification Layer

To combine explainability with predictive accuracy, the FastAPI backend first applies interpretable, weighted formulas to compute a Vulnerability Score, based on the relative share of senior citizens, PWDs, pregnant women, and children, and a Priority/Hazard Score, based on flood depth and the barangay's vulnerability modifier. These scores are surfaced on the frontend as an explainable basis for each classification. The backend then feeds the underlying features, together with encoded flood depth category, flood status, and cause of flooding, into trained Random Forest and XGBoost classifiers — used comparatively to support accuracy in priority classification — to output the final, city-wide Priority Level (High, Medium, or Low) for each registered street. These automated predictive classifications are seamlessly surfaced on the frontend Priority Ranking Dashboard, allowing DRRM Officers to see an evidence-based, model-driven view of where preparation and evacuation resources must be directed first.

3. Incident Verification, Mapping, and Operations Layer

To maintain high data integrity, LIKAS implements a dual-role incident verification workflow. Barangay accounts can report active street inundations through a map-based, multi-step Log Incident flow, complete with photo attachments (required for barangay users, optional for admin), which enter the Flood Incident Management module under a Pending state. Admin accounts maintain exclusive clearance to review incoming field logs, verify the attached incident evidence, and switch the status to Approved or Rejected. Once approved, the updated features are instantly reprocessed by the scoring engine, dynamically re-classifying the affected area's priority ranking. The Priority Ranking Dashboard further visualizes these results on an interactive Leaflet-based map of Manila City, using official Philippine administrative boundary data, alongside dashboard analytics of flood trends, causes, and priority distribution, with report export supported through PDF generation.

The system will be evaluated across three primary dimensions: the predictive performance of the Random Forest and XGBoost classifiers (accuracy, precision, recall, and F1-score), system load and performance stability, and user experience metrics evaluated with barangay and MDRRMO stakeholders.

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